

CS 631 - Project

Deliverable 3

Requirements

Phase 3 must include the goals of this phase, a description of the creation of the database instance, and the development of application programs. It should also document any revisions made to the relational schema presented in Deliverable 2, along with challenges encountered and the rationale behind your solutions.

- The program must run
 - Make sure you have ample sample data in your tables so that you can sufficiently perform and illustrate all required tasks.
 - The user interface should be menu-driven and practical. Do not spend time developing a sophisticated GUI unless you have extra time.
1. Write SQL command files that populate each table. Each table is required to have a sufficient number of tuples. Keep in mind the relationships between tables and plan your data accordingly.
 2. You are to develop a menu driven application system for the Research Lab Manager database. The following are the menus to be developed. All applications described below must be implemented. You may add additional features if desired.

1. PROJECT AND MEMBER MANAGEMENT

- Query, add, update, and remove members and projects.
- Display the status of a project.
- Show members who have worked on projects funded by a given grant.
- Show mentorship relations among members who have worked on the same project.

2. EQUIPEMENT USAGE TRACKING

- Query, add, update, and remove equipment and equipment usage.
- Show status of a piece of equipment

- Show members currently using a given piece of equipment and the projects they are working on.

3. GRANT AND PUBLICATION REPORTING.

- List the top 5 projects ranked by their total grant funding, and show the total amount each project received, in decreasing order of total funding (assume Budget is the dollar amount of each grant).
- Find the mentor(s) whose mentees collectively produced the largest number of publications.
- Calculate the total number of student publications per major and per publication year.
- Given a date X, find the projects that ended before X and the number of grants that funded each project.
- Find the three most productive years in terms of publications produced by students.

You will be required to demonstrate your project at the end of the semester. A reservation form will be made available online to schedule your demo time slot.